

Regulation of Transcriptional Bursting and Spatial Patterning in Early *Drosophila* Embryo Development

Reviewed Preprint

v1 • September 23, 2025

Not revised

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eLife Assessment

This study presents a **valuable** quantitative framework for analyzing transcription dynamics data for enhancers and genes expressed in the early *Drosophila* embryo. By analyzing existing data across both synthetic reporters and an endogenous gene (*eve*), this work provides evidence that spatial gene expression patterns within the embryo are largely determined by "activity time" - the time during which a gene is bursting. The methods and evidence are **solid** and should be of broad interest to researchers in developmental biology and quantitative gene regulation, but the study would be significantly enhanced by clarifying the novelty of the findings relative to prior work and presenting a rigorous benchmarking of their algorithm against previously used algorithms.

<https://doi.org/10.7554/eLife.107610.1.sa4>

Abstract

Nascent RNA synthesis often occurs in periods of high transcriptional activity, interspersed with basal or no activity periods. This phenomenon, known as transcriptional bursting, drives high intercellular variability in gene expression levels. How do key patterning genes in early *Drosophila melanogaster* embryos overcome this variability to establish precise spatial patterns for tissue development? To address this question, we study single cell transcriptional activity from MS2-based live imaging data of four transgenic constructs (*rho*, *Kr*, *sna* shadow, *sna* proximal enhancers) and the endogenous *eve* gene. We developed an algorithm to infer promoter states in hundreds of cells within the embryo using transcriptional activity data. Results show that while mean transcription levels exhibit spatial gradients, the burst duration and interburst timing remain surprisingly invariant across the embryo and different constructs. The time between consecutive bursts was consistent with a memoryless exponential distribution, whereas the burst duration exhibited tighter control with lesser stochastic variations. Our analysis identified two regulatory mechanisms for gene expression

gradients: (1) similar burst-timing statistics across genes, with the activity time (the time from the first to the last burst) being modulated to regulate distinct expression levels; (2) for the same gene with different enhancers (*sna* shadow and *sna* proximal), we observed changes in the mean burst duration and variability of the inter-burst timing. This study provides a comprehensive approach to analyzing transcriptional bursting kinetics, revealing activity time as a major regulator of spatiotemporal expression patterning in early embryonic development.

Introduction

Robust regulation of gene expression is essential for normal cell physiology and embryo development. Variations in expression levels or ectopic activation at inappropriate times or locations can significantly affect cell phenotypes, fitness, survival, and differentiation [1–3]. Gene misregulation can be related to different developmental defects and disease phenotypes [4, 5]. Despite the need for strict control, single cell imaging has revealed random fluctuations (noise) in promoter transcriptional activity over time [6, 7]. Several studies have characterized transcriptional noise in the expression of developmental genes across organisms [8–15], and it is important to understand how this noise in single cells affects overall spatial expression patterning and subsequent development.

Gene expression is characterized by transcriptional bursts: intermittent periods of promoter activity followed by periods of inactivity in which transcripts are not produced [16–26]. Randomness in switching between active and inactive promoter states has been shown to be a key driver of intercellular variability in gene product levels across isogenic cell populations [27–37]. By analyzing the properties of transcriptional bursting in various genetic backgrounds, we can better understand the mechanisms for regulating the levels of gene products [16, 38, 39].

We present a comprehensive analysis of the transcriptional bursting properties of reporter genes driven by various regulatory enhancers and endogenous genes in early *Drosophila* embryos. We analyze publicly available live imaging data where single-cell transcriptional activity is tracked as a function of time and position in the embryo [40, 41]. Each gene is labeled with the 24x MS2 sequence, which allows visualization of nascent transcripts in each cell with a fluorescent marker (MCP-GFP), allowing high temporal resolution to investigate the dynamics of bursting [42, 43]. We develop a simple yet robust algorithm to estimate promoter transcriptional states based on the accumulation and decay of individual fluorescence trajectories. These inferred promoter states are then used to quantify the spatiotemporal statistics of transcriptional bursting across the gene expression domain.

Among the properties of transcriptional bursting studied, we consider the burst duration (ON time), the time between successive bursts (OFF time), the rate of nascent mRNA transcription (loading rate), and the span from the first to the last burst (activity time). To examine how parameters such as spatial heterogeneity, different genes, or different enhancers affect burst dynamics, we selected enhancers of *Krüppel* (*Kr*) and *rhomboid* (*rho*) genes. The mean transcriptional activity of these genes drives expression that follows a graded pattern along the anterior-posterior and dorsoventral axes of an embryo, respectively [44, 45]. Interestingly, we find that the mean ON and OFF times are homogeneous across the expression domain, not showing the expression gradient observed in the average transcription activity. This heterogeneity can be mainly explained by differences in activity time, suggesting that different cells have different durations of active transcription but similar bursting properties once activated.

We compare two redundant *sna* enhancers (*sna* shadow and *sna* proximal), which drive spatially homogeneous ventral expression, revealing distinct bursting characteristics and indicating that enhancers can modulate burst timing statistics [46]. Furthermore, we study the endogenous *even-skipped* (*eve*) gene, with its seven-strip spatial pattern [47, 48]. Although *eve* shows homogeneous burst duration, we also found spatially varied activity time and inter-burst duration, aligning with observations from the *Kr* and *rho* constructs.

Results

This study characterizes the spatiotemporal dynamics of key genes involved in *Drosophila* embryonic development. We analyze gene expression dynamics during the 14th nuclear cycle (NC14), which corresponds approximately to a time window of 2-3 hours after fertilization. We use the enhancers of *rho*, *Kr*, and *sna*, which drive transcription of the MS2 reporter genes that recapitulate endogenous gene expression [21]. We also analyze the transcription of the endogenous *eve* gene (see Methods). All of these genes play an important role in the formation of body patterns in *Drosophila* [49].

Spatio-temporal properties of transcription bursts

The experiments analyzed (see Methods) consist of confocal microscope images of living *Drosophila* embryos (Figure 1A). These images discern fluorescence dynamics from individual cells, allowing single-cell resolution analysis along the anterior-posterior and dorsoventral axes of the embryo (Figure 1B). This fluorescence trajectory corresponds to transcription dynamics captured using the MS2/MCP system (Figure 1C). In this system, the number of nascent mRNA molecules is proportional to the fluorescence signal (Figure 1D, top panel). This trajectory reveals different periods of intense activity (referred to as bursts) with stochastic duration indicated by τ_{ON} and intervals between successive bursts denoted as τ_{OFF} . To estimate the transcriptional state, we developed an algorithm (see Methods) based on the finding that the fluorescence signal increase (during ON state) and decrease (during OFF state) is approximately linear over time (Figure S1), with relatively similar slopes (Figure S2). In Figure 1D bottom, we show an example of an inferred transcription state trajectory from the fluorescence signal (see Supplementary Information Section S3 and Figure S3 for technical details of the inference algorithm).

The fluorescence intensity averaged throughout the nuclear cycle reveals a distinct spatial pattern in the embryo (Figure 1A). Here, we relate this mean fluorescence to the statistics of τ_{ON} and τ_{OFF} . In addition to τ_{ON} and τ_{OFF} , our algorithm also estimates the *loading rate* λ^* , which is defined as the signal increase rate that best predicts the signal dynamics (see Methods) and is specific for each cell. Finally, we also estimate the time between the beginning of the first burst and the end of the last burst, which is called the *activity time*. We will study how these different burst properties vary spatially across the embryo, and which properties have the best concordance with the single-cell averaged mean fluorescence level.

The activity time as an important contributor to the control of gene expression

We first analyze the burst statistics of the embryos expressing *rho* and *Kr* genes. *rho* shows a graded expression along the dorsoventral (vertical) axis and *Kr* shows a gradient along the anterior-posterior (horizontal) axis (see Figure 2A and 2G) [44, 45]. *rho*NEE enhancer that drives the neuroectodermal expression of *rho* is used to drive the reporter gene MS2-yellow expression [45]. On the other hand, the expression pattern of *Kr* is driven by the enhancer *Kr* CD2 [50].

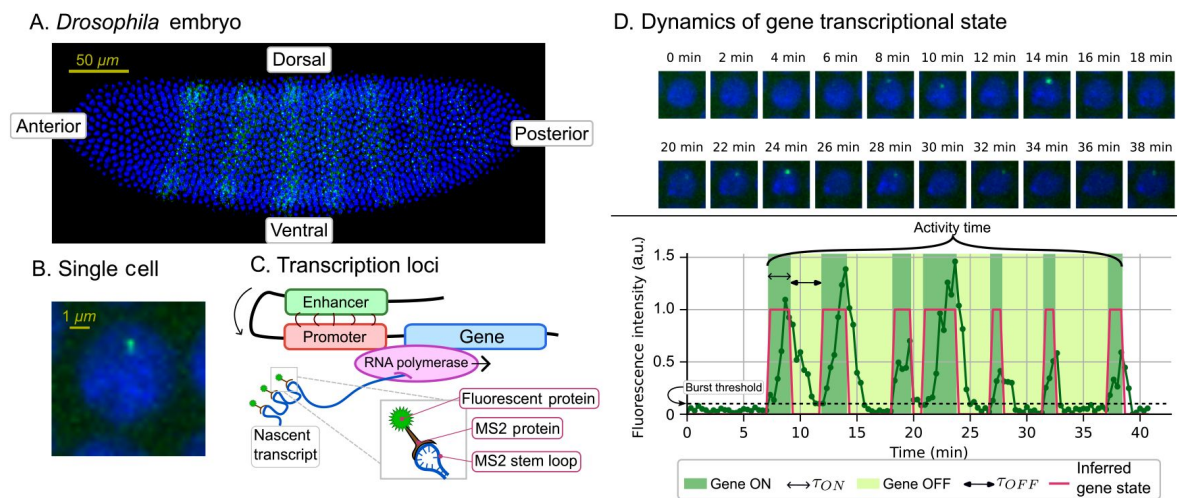


Figure 1

The MCP/MS2-mediated live imaging system enables the visualization of transcriptional activity at the single-cell level in living *Drosophila* embryos.

(A) Representative snapshot of MS2-based imaging, where endogenous *eve* is tagged with MS2 to visualize the spatial expression pattern in embryos. Regions of the embryo (ventral, dorsal, anterior and posterior) are provided. (B) A single cell in the embryo showing GFP fluorescence with intensity assumed to be proportional to the number of nascent mRNAs. (C) Schematic showing how the nascent transcripts are tagged with fluorescent proteins using the MS2/MCP system. (D) (Top:) Representative snapshots of a single nucleus producing transcripts (green puncta) over time. (Bottom:) Fluorescence dynamics over time (green line) and the inferred transcriptional state (red line). From the inferred state, we estimate the duration of active (ON) states (τ_{ON}) and the duration of inactive (OFF) states (τ_{OFF}). The black dashed line shows the burst threshold: the level where any sudden increase from the base level is considered as a burst (see Methods). Bracket in the top of the trajectory represents the activity time: the time between the beginning of the first burst and the end of the last one.

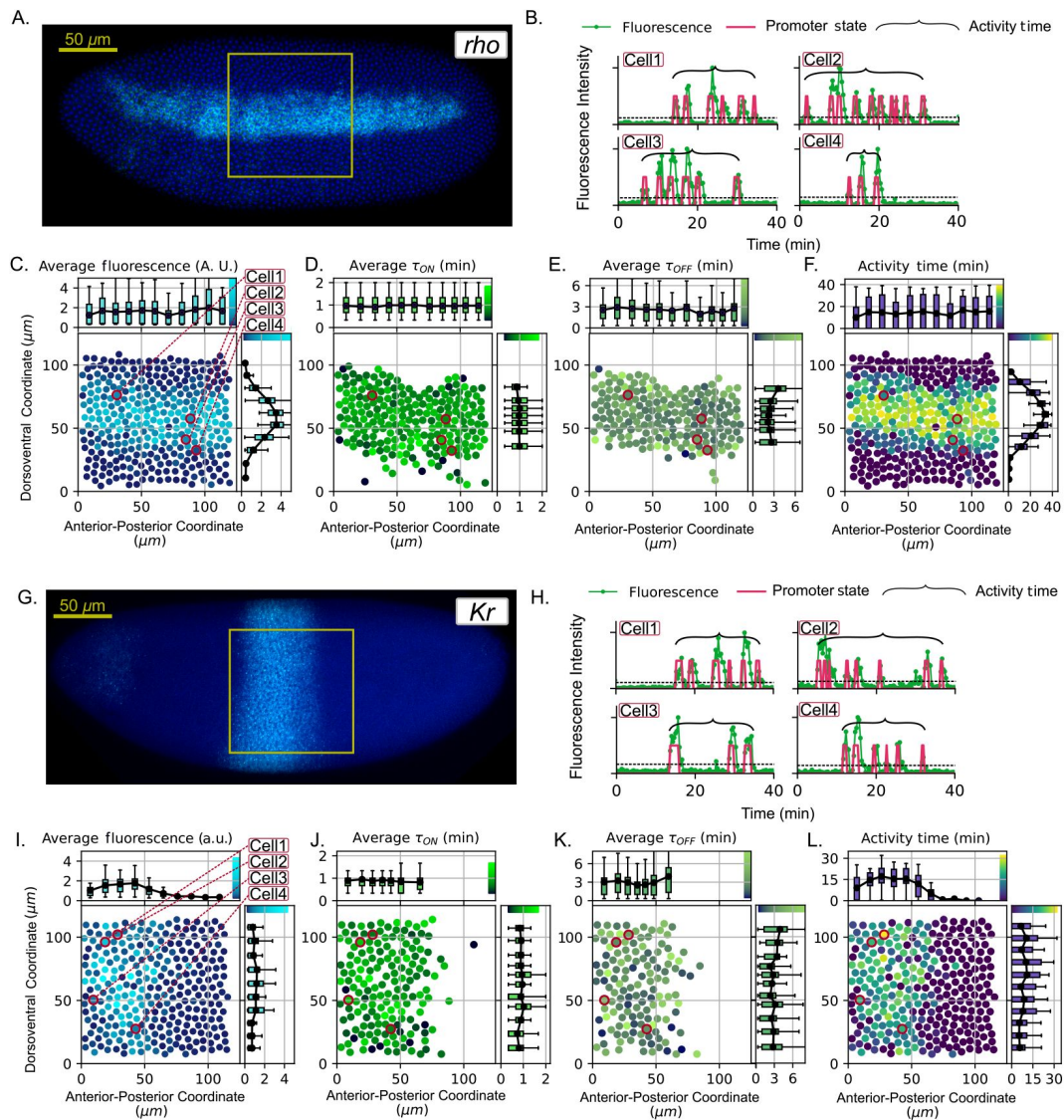


Figure 2

Heterogeneous spatial expression patterns for enhancer *rho* and *Kr*.

(A,G) Gene expression driven by the *rho* (A) and *Kr* (G) enhancers which exhibit a dorsoventral and anterior-posterior gradient, respectively. The yellow box indicates, approximately, the region imaged by live imaging. (B,H) Representative fluorescence trajectories (green) and the inferred transcriptional state (red) for *rho* (B) and *Kr* (G). Fluorescence signals are normalized to their maximum. Activity time is represented by black brackets. Dashed horizontal line represents the burst threshold. (C,I) Average fluorescence intensity, (D,J) cell averaged τ_{ON} , (E,K) average τ_{OFF} , and (F,L) the activity time. The position of each point represents one cell in the beginning of NC14 and its color represents the mentioned average burst property. Cells presented in (B,H) are highlighted in red in (C-F) and (I-L) respectively. Marginal plots show the mean variables calculated with a spatial binning. The error bars show the 5% and 95% quantiles, while the length of the boxes show the 25% and 75% quantiles for each bin. Black dots represent the mean value. The means are connected by lines to show the spatial trends. Marginal plots also present the color scale at their edges used in the central scatter plot.

Our initial hypothesis was that gene expression can be directly associated with the frequency and duration of transcriptional bursts. This means that in regions with low or high mean gene expression, we expect less or more frequent bursts, respectively. To verify the validity of this hypothesis, we plot the spatial patterns of the average fluorescence intensity of *rho* (**Figure 2C-F**). As examples of observed dynamics, four representative nuclei from the gene expression domain were selected and their transcriptional trajectories were plotted (**Figure 2B**). In **Figure 2C**, we show a scatter plot with each dot representing one cell. The position of each dot represents the position of the respective tracked cell at the beginning of NC14. The color of the dots represents the fluorescence level averaged throughout NC14. To better visualize the spatial gradient of the burst properties, we also made marginal plots by grouping the cells into bins along the dorsoventral and anterior-posterior axis and calculating the statistics for each bin.

Figure 2D shows spatial patterns of the cell-averaged τ_{ON} . We excluded cells that did not show fluorescence levels above the burst threshold, as it is not possible to estimate the average τ_{ON} . The marginal plots in **Figure 2D** show the variation of the cell-averaged τ_{ON} for the selected bins. These bins reveal that the mean τ_{ON} for multiple regions of the embryo is constant, with a value of about 1 min. Similar spatial trends for τ_{OFF} are presented in **Figure 2E**. Marginal plots show that the mean τ_{OFF} along the axes also shows a low variation throughout the embryo. In this case, the average τ_{OFF} is around 3 min.

The finding that the cell-averaged τ_{ON} and τ_{OFF} is spatially uniform contradicts the hypothesis of a non-homogeneous gene switching rate. To explain pattern formation in the selected genes, we noticed, as presented in **Figure 2B**, that the dynamics of gene expression can be simplified as periods of activity. During these long periods, multiple bursts of similar duration occur at an approximately constant rate. This motivated us to define the activity time as the span from the first burst to the final one. The duration of this activity time is marked with a bracket on the signal trajectory in **Figure 2B**.

We notice that spatial gene expression corresponds better to the patterns of activity time (**Figure 2F**) than to the cell-averaged τ_{ON} and τ_{OFF} . This suggests that the activity time better reflects the spatial heterogeneity of gene expression. This is confirmed by the *Kr* construct, where mean fluorescence shows a similar spatial pattern to activity time (**Figures 2G-2L**). Although *Kr* has different mean fluorescence levels along the embryo axis compared to *rho*, the average τ_{ON} and τ_{OFF} in *Kr* are relatively homogeneous with mean durations similar to those in *rho*. Our analysis indicates that the graded expression pattern results from the modulation of activity time rather than bursting properties.

Different enhancers can modify the time between successive bursts

Enhancers are short non-coding DNA sequences that interact with the target promoter to drive transcription (**Figure 1C**). Particular enhancers can increase the likelihood of transcriptional initiation by containing binding sites for transcription factors and other proteins that promote transcription [51]. We now ask whether different enhancers of the same gene can affect the burst-timing statistics. We studied the *sna* gene, which has a broad expression domain on the ventral side of an embryo, as shown in **Figure 3A**. *sna* is regulated by two enhancers, the proximal enhancer (*snaPE*) that is located right upstream of the transcription start site (TSS) and the distal or shadow enhancer (*snaSE*) located 8kb upstream of the TSS [46]. Both enhancers are known to drive *sna* expression in the same domain at the same time, but *snaPE* drives weaker expression than *snaSE* (**Figures 3B** and **3G**) [52].

Comparing the bursting properties of *snaPE* and *snaSE*, we find some similarities with the *rho* and *Kr* genes. For example, τ_{ON} and τ_{OFF} have low spatial heterogeneity and the activity time is highly correlated with the average fluorescence (**Figures 3C-F** and **3H-K**). However, some

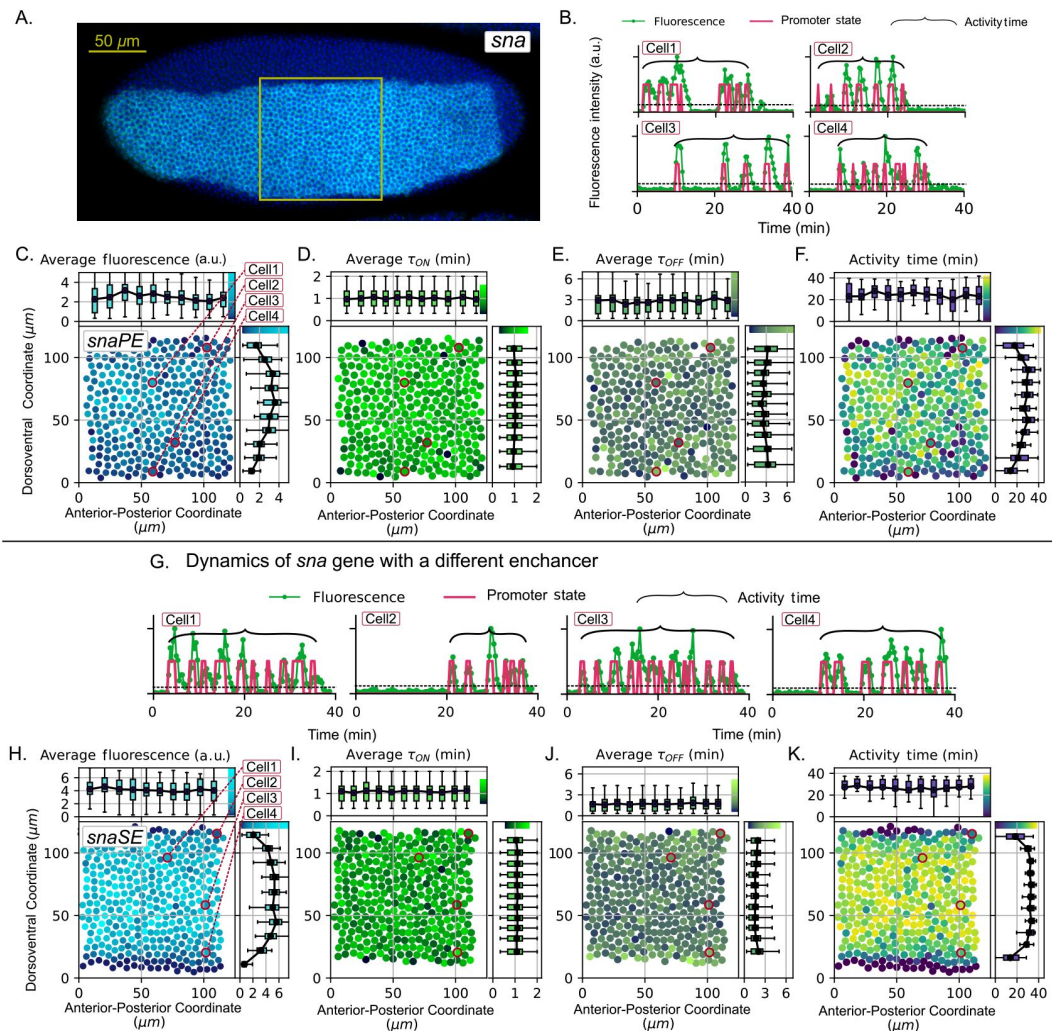


Figure 3

Spatial patterns of *sna* gene expression for different enhancers: shadow (SE) and proximal (PE)

(A) Image of the *sna* expression across the embryo. The yellow box indicates, approximately, the region imaged. (B, G) Representative gene expression trajectories normalized to the maximum (green) and the inferred transcriptional state (red) for *sna*PE (B) and *sna*SE (G). Activity time is represented by black brackets. Dashed horizontal line represents the burst threshold. These cells are highlighted in red in (C-F) and (H-K). (C, H) Spatial trends of average fluorescence, (D, I) average τ_{ON} , (E, J) average τ_{OFF} , and (F, L) activity time. Marginal plots show how the mean changes in different spatial bins of the expression domain. Bar length represents the 25-75% interquartile distance and error bars, the 5-95% interquartile distance.

differences are also observed. For example, the mean fluorescence intensity is higher for the SE enhancer than for the PE enhancer, although both regulate the same gene (Figures 3C and 3H). This difference can be explained by a shorter average τ_{OFF} for the SE enhancer. Since *snaSE* is considered a stronger enhancer than *snaPE*, producing more mRNA at a given duration, our results suggest that *snaSE* can drive stronger expression by spending less time in the OFF state, which can also explain similar results found previously [40]. Our analysis indicates that different enhancers of the same gene can induce different bursting behaviors and therefore regulate the level of mRNA production differently.

Statistical analysis of transcriptional bursting events

To better understand the statistical properties of the burst timing variables, we collect all the values of τ_{ON} and τ_{OFF} for different replicate embryos and study their statistical distribution without discrimination by cell or position. More specifically, we quantify the mean τ_{ON} and τ_{OFF} (denoted by $\langle \tau_{ON} \rangle$ and $\langle \tau_{OFF} \rangle$, respectively). The random variability in τ_{ON} and τ_{OFF} is measured by the squared coefficient of variation (the variance over the mean squared) and represented by $CV_{\tau_{ON}}^2$ and $CV_{\tau_{OFF}}^2$, respectively. These statistics are estimated over all cells in the embryo during NC14 for 3 different embryos (biological replicas) per construct.

While τ_{OFF} follows an exponential distribution, τ_{ON} has a hypo-exponential distribution

The analysis of $\langle \tau_{ON} \rangle$ and $\langle \tau_{OFF} \rangle$ is presented in Figure 4A. There, we show that $\langle \tau_{ON} \rangle \approx 1$ min while $\langle \tau_{OFF} \rangle \approx 3$ min for *Kr*, *rho*, and *snaPE*. The main outlier corresponds to *snaSE* ($\langle \tau_{OFF} \rangle \approx 2$ min). These results show that the properties of the burst dynamics are consistent not only for different replicas, but also among different genes. Regarding the random variability of the time intervals, Figure 4B shows how while $CV_{\tau_{ON}}^2 \approx 0.2$ in all replicas and genes, $CV_{\tau_{OFF}}^2 \approx 1$ for *Kr*, *rho*, and *snaPE*. The main outlier, *snaSE*, shows a variability of $CV_{\tau_{OFF}}^2 \approx 0.5$. As a reference for the interpretation of this metric, we recall that the coefficient of variation for an exponentially-distributed random variation is 1 (dashed line in Figure 4B).

To better visualize the interpretation of the metrics of $CV_{\tau_{ON}}^2$ and $CV_{\tau_{OFF}}^2$, we plot the histograms of τ_{ON} for *snaPE* and *snaSE* (Figure 4C). For these enhancers $CV_{\tau_{ON}}^2 \approx 0.24 < 1$. This means that the distribution of τ_{ON} is hypoexponential, i.e., with less random variation than an exponential distribution (which has a coefficient of variation of 1). We also plot the histogram of τ_{OFF} in Figure 4D for the same enhancers. τ_{OFF} for *snaPE* has a mean of approximately 2.8 min and a noise of $CV_{\tau_{OFF}}^2 \approx 1$ similar to the exponential distribution. *snaSE*, on the contrary, shows a shorter $\langle \tau_{OFF} \rangle \approx 1.7$ min and a lower variability $CV_{\tau_{OFF}}^2 \approx 0.5$.

Correlation between burst variables and mean fluorescence

In this section, we explicitly examine how the cell-averaged fluorescence relates to four other variables: cell-averaged τ_{ON} , cell-averaged τ_{OFF} , activity time and optimal loading rate λ^* (equation (3) in Methods). We gather data from three biological replicate embryos for each construct and observe the trends between these variables.

We observe that activity time increases monotonically with increasing mean fluorescence intensity, reaching saturation at high levels of mean fluorescence (Figure 5A). On the other hand, cell-averaged τ_{ON} , cell-averaged τ_{OFF} , and λ^* show weaker correlations with mean fluorescence. Cell-averaged τ_{ON} increases slightly with increasing mean fluorescence, validating our simplification that τ_{ON} is essentially constant across transgenic constructs (Figure 5B). The cell-averaged τ_{OFF} shows a slightly decreasing trend with mean fluorescence, although the trend is

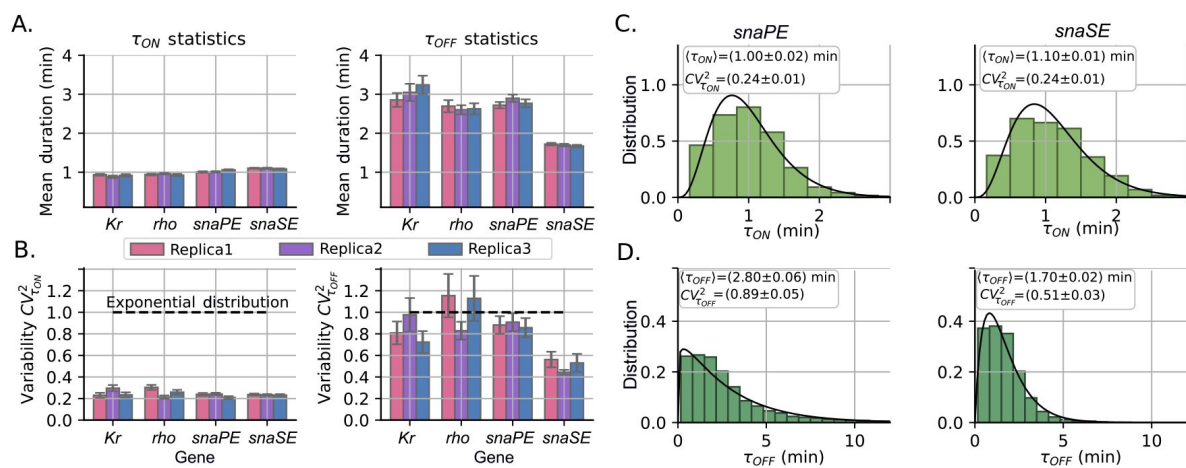


Figure 4

Statistical properties of transcriptional bursting timing across different constructs.

(A) Mean duration of τ_{ON} and τ_{OFF} for the four studied enhancers, each with three replicates. (B) Random variability of τ_{ON} and τ_{OFF} measured by the squared coefficient of variation $CV^2_{\tau_{ON}}$ and $CV^2_{\tau_{OFF}}$ respectively. The dashed line represents the coefficient of variation of one for an exponentially-distributed random variable. (C) Probability density for τ_{ON} for the *sna* enhancers. The PE enhancer (left) is compared to the SE enhancer (right). The mean and squared coefficients of variation are presented with the 95% confidence interval. The solid line represents the best fit to a gamma distribution. (D) Analysis similar to (C) but for τ_{OFF} . The resolution of τ_{ON} and τ_{OFF} is 20 s.

weaker compared to the relationship between activity time and mean fluorescence (**Figure 5C**). Relatively longer periods of τ_{OFF} are observed for nuclei with lower gene expression. Finally, there is almost no correlation between the loading rate λ^* and the mean fluorescence (**Figure 5D**).

The statistics of the *eve* gene expression

Until now, the data analyzed to estimate the burst statistics correspond to transgenic constructs. Next, we performed a similar analysis of the transcriptional burst statistics of *eve*, an endogenous gene with a more complex spatial pattern of expression. The average fluorescence level of *eve* has a pattern of seven stripes during early embryonic development (**Figure 6A**) [47]. These stripes are the result of interactions between *eve* and gap genes defining the position of specific para-segments that set the stages for the final segmentation of the embryo body [53]. As observed in **Figure 6A**, we analyze transcription trajectories of the nuclei corresponding to stripes 2-5. Examples of typical fluorescence trajectories of *eve* expression can be seen in **Figure 6B** showing, in general, a lower average intensity than the synthetic constructs (*Kr*, *rho* and *sna*) but a similar basal intensity. Regarding the timing statistics, we observe low spatial variability on the cell-averaged τ_{ON} and τ_{OFF} on the spatial domain of *eve* (**Figure 6D** and **6E**). Furthermore, these mean times also show values similar to those of the synthetic constructs: $\langle \tau_{ON} \rangle \approx 1$ min and $\langle \tau_{OFF} \rangle \approx 3$ min. Again, the activity time appears to align better with the average fluorescence pattern (**Figure 6F**).

Figure 7 presents a more detailed analysis of the relationship between mean fluorescence and burst properties for the *eve* gene, based on data from six biological replicates. As with synthetic constructs, cell-averaged fluorescence shows a strong alignment with activity time (**Figure 7A**), but only a weak correlation with cell-averaged τ_{ON} (**Figure 7B**) and loading rate (**Figure 7D**). Cell-averaged τ_{OFF} shows a stronger decrease with mean fluorescence compared to the one observed in synthetic constructs (**Figure 7C**). The mean burst duration for *eve* shows relative consistency over replicas with similar mean (**Figure 7E**) and $CV_{\tau_{ON}}^2 \approx 0.25$ (**Figure 7F**). However, τ_{OFF} has slightly less consistency on the mean $\langle \tau_{OFF} \rangle$ while the random variability $CV_{\tau_{OFF}}^2$ was hyperexponential ($CV_{\tau_{OFF}}^2 \approx 2.5 > 1$).

The relatively large $CV_{\tau_{OFF}}^2$ observed in *eve* is mainly due to the presence of abnormally long inter-burst times, often exceeding 12 min (Figure S4A). To validate this concept, we reanalyzed the burst statistics, focusing exclusively on cells with inter-burst times of less than 12 min. After excluding these cells, all burst properties align with those observed in synthetic constructs: $\langle \tau_{ON} \rangle \approx 1$ min and $\langle \tau_{OFF} \rangle \approx 3$ min, while $CV_{\tau_{ON}}^2 \approx 0.25$ and $CV_{\tau_{OFF}}^2 \approx 1$ (Figure S4C-D). The comparison of the burst properties versus the mean fluorescence after this filtration is also presented in Figure S4E showing a clearer trend in activity time versus fluorescence and a flatter relationship between the mean τ_{OFF} and the mean fluorescence. We believe that these long periods are due to the *eve* regulation by multiple stripe-specific enhancers, in contrast to transgenic constructs, which typically correspond to the effects of a single enhancer. The complex interaction between these enhancers likely contributes to increased time variability while maintaining a similar spatial pattern of $\langle \tau_{ON} \rangle$ and $\langle \tau_{OFF} \rangle$ across stripes. In addition, it is known that *eve* stripe expression shifts anteriorly during NC14, such that some cells expressing *eve* at the posterior boundary of a stripe early in NC14 become transcriptionally inactive later [41]. For example, a cell located between stripe 1 and stripe 2 may initially express genes early in NC14 due to the activity of stripe 1 enhancers. Midway through NC14, it becomes transcriptionally inactive and later in NC14, as the expression domain shifts, it begins to express genes regulated by stripe 2-specific enhancers. This trajectory results in a long inactivity period between bursts, which contributes to higher variability. In fact, nuclei with inter-burst times of more than 12 min are mostly located at the edges of each stripe (Figure S4B).

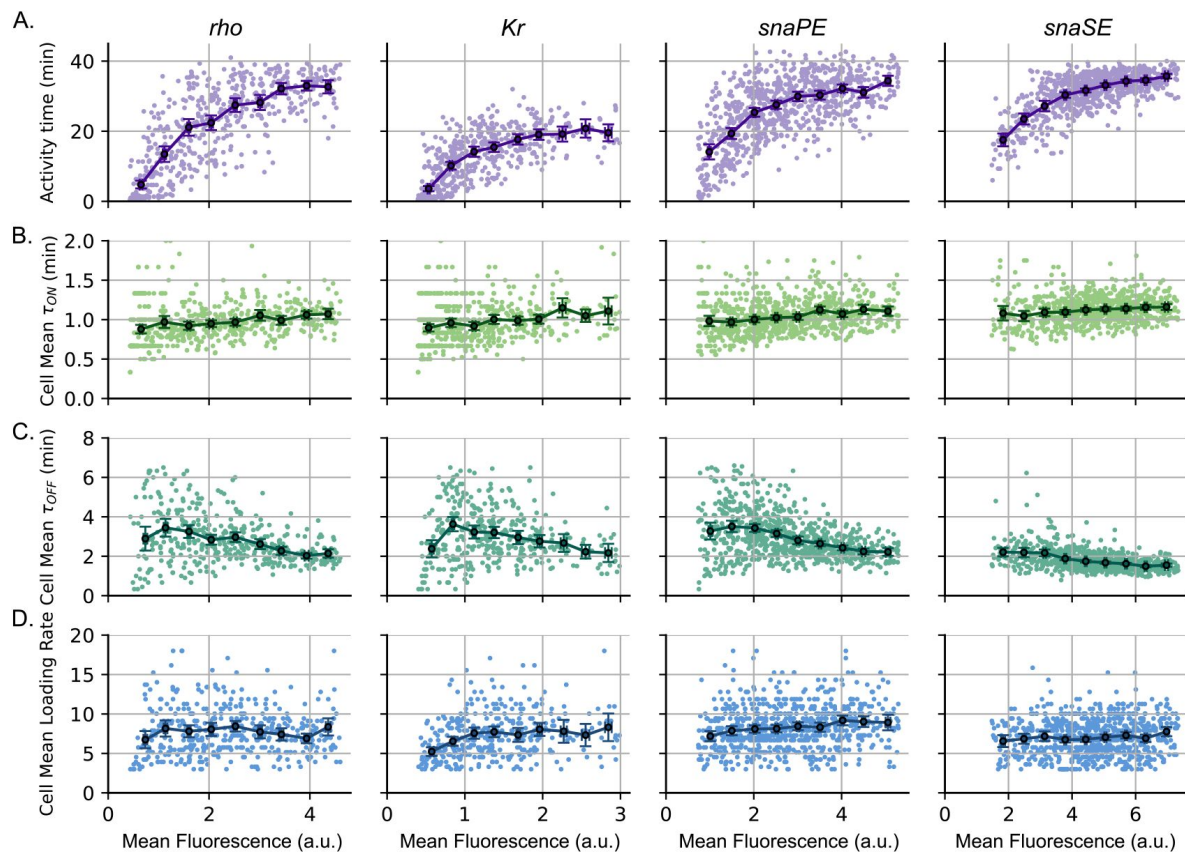


Figure 5

Statistical properties of the transcriptional bursts with respect to the mean fluorescence for different enhancers.

We plot the trend of different signal statistics such as **(A)** activity time, **(B)** mean ON state duration, **(C)** mean OFF state duration, and **(D)** mean loading rate (the rate of signal increase) relative to the mean fluorescence. Each dot in the plots represents the mean value of a single cell obtained from its fluorescence trajectory during NC14. These trends suggest that the activity time aligns better with mean fluorescence.

Figure 6

Spatial patterns of endogenous *eve* gene expression.

(A) Expression pattern of the *eve* gene. The yellow box indicates, approximately, the region imaged. (B) Expression trajectories normalized to the maximum (green) and the state of the inferred gene (red). Activity time is represented by black brackets. The horizontal dashed line represents the burst threshold which is the same as used in the constructs. These cells are highlighted in red in (C-F). (C) Spatial trends of average fluorescence, (D) average τ_{ON} , (E) average τ_{OFF} , and (F) the activity time in the studied embryo domain. Bar length represents the 25-75% interquartile distance and error bars, the 5-95% interquartile distance.

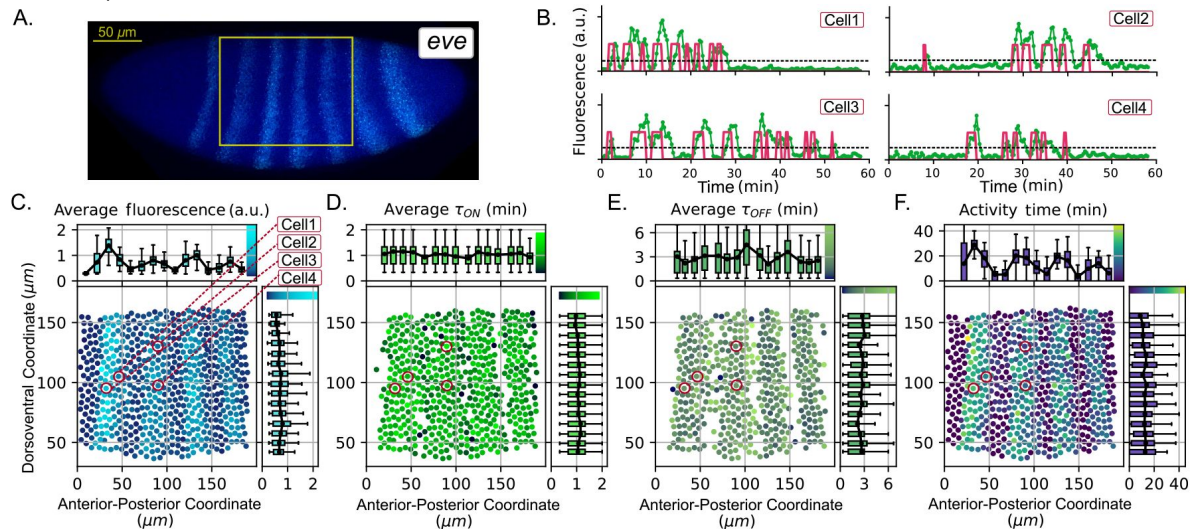
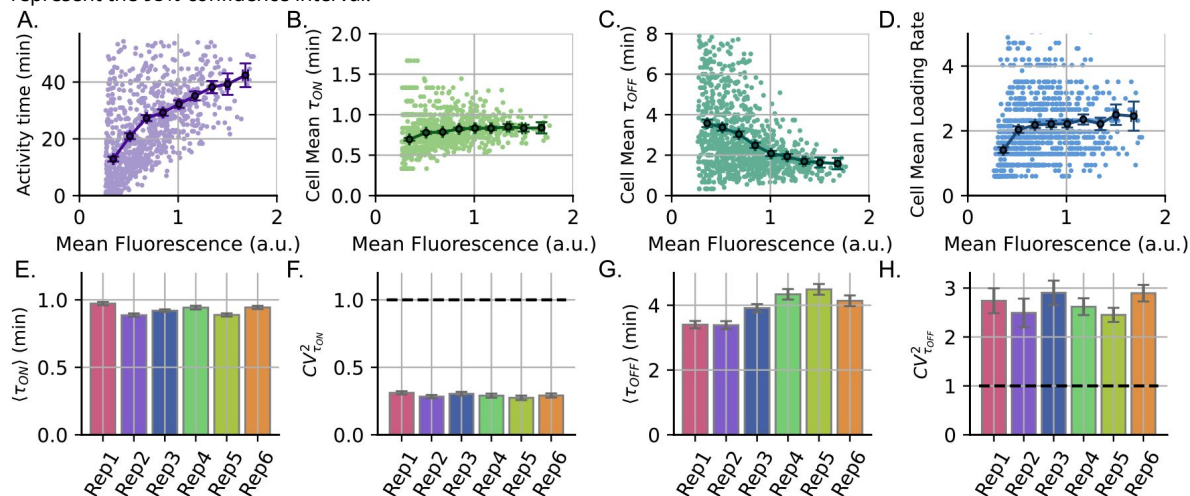


Figure 7

Properties of transcriptional burst timing for the expression of endogenous *eve* gene.

Plots of average burst properties with respect to single-cell averaged fluorescence level (A) activity time, (B) Mean τ_{ON} , (C) Mean τ_{OFF} and (D) Mean loading rate. Each dot in these plots corresponds to a cell and the statistics are calculated over the NC14 time-span. Error bars represent the binned mean trend with 95% confidence. We also present the statistics of the burst timing estimated using all bursts and all cells for 6 different biological replicas: (E) mean burst duration (τ_{ON}), (F) variability on burst duration $CV^2_{\tau_{ON}}$, (G) mean time between successive bursts (τ_{OFF}) and (H) variability in the time between bursts $CV^2_{\tau_{OFF}}$. Horizontal dashed lines in (F) and (H) correspond to a coefficient of variation of 1 for the exponential distribution. Error bars represent the 95% confidence interval.



In summary, our analysis suggests that the bursting properties of endogenous genes are comparable to those of developmental enhancers. These findings align with a previous study that characterized the bursting properties of *eve* using a reporter construct based on bacterial artificial chromosomes [48, 54]. Similarly to what has been observed in enhancers, the mean gene expression level for the *eve* gene is mainly aligned with changes in activity time despite showing a more complex spatial pattern (Figure 7A). We observed some cells with anomaly long interburst duration that biased the τ_{OFF} statistics. We hypothesized that these long periods are likely due to the coordinated action of multiple enhancers that collectively regulate *eve* expression.

Discussion

In this article, we aim to explain how the spatial pattern of mean gene expression in the early embryonic development of *Drosophila* emerges from the temporal dynamics of transcriptional burst activity at the single cell level. We analyze the dynamics of the amount of nascent RNA molecules measured using MS2 tagging. To infer the transcriptional state (active or inactive), we propose an algorithm based on the signal slope (see Methods). The simplicity of our algorithm helps us analyze thousands of fluorescence trajectories from individual nuclei with high throughput, providing reproducible results on inferred transcriptional states. Our analysis reveals that fluorescence accumulation and decay follow linear functions of time (Figure S1). Although simple, this dynamics requires a careful explanation using a model that considers a detailed description of RNA polymerase recruitment and elongation on the gene. Although this kind of modeling is beyond the scope of this article, we show that the process can be simplified by a two-state model as described by equations (1) and (2) in Methods to capture key transcriptional burst properties.

On the basis of the statistics of transcriptional bursts, it is possible to infer properties of the mechanism controlling the burst timing. There are known mechanisms that drive and regulate transcriptional bursting, such as chromatin dynamics [55–59], 3D enhancer-promoter interactions [54, 60–63], transcription factor binding [64–68], mRNA polymerase pausing [69], and interactions between developmental promoter motifs and associated proteins [70]. Our observation of exponentially-distributed interburst timing (Figure 4B) suggests that burst initiation may be governed by a single rate-limiting step in these developmental enhancers. The observation of burst-duration timing to be hypoexponentially distributed as shown in Figure 4B (that is, these times have statistical fluctuations less than an exponential distribution) implies a more complex transcriptional turn-OFF with multiple rate-limited steps. Findings of non-exponential promoter state durations have previously been made for other promoters in their OFF times [71, 72], and also for their ON times [73]. Further analysis of whether the durations of consecutive bursts are related (Figure S5) revealed weak correlations suggesting ON/OFF times being independent and identically distributed random variables as good approximations for the observed bursting behavior.

There is a rich body of literature connecting transcriptional bursting to fluctuations in levels of a given gene product [74–84]. Borrowing this recent mathematical modeling framework, we explore the impact of non-exponential promoter switching times on the statistical fluctuations in gene expression levels. We linked transcription states (ON/OFF) to the generation and degradation of a gene product, with the time spent in each state itself being an independent and identically distributed random variable following an arbitrary distribution (see Supplementary Information Section S5). Our analysis shows that the noise in mRNA levels increases with the noise in the time spent in either of the ON and OFF states (Figure S6). These results imply that tight control of transcriptional burst durations, as seen in this study, will function to buffer stochastic variations in expressed mRNA levels. An intriguing finding is the consistency in the statistics of promoter ON/OFF times across cells and different genes suggesting common mechanisms controlling bursting timing. It is worth mentioning that many studies have suggested that the kinetics of

bursting can vary under different conditions [48, 85, 86], and that the time between successive bursts was observed to be controlled by the gene enhancer. This control includes, as shown here, decreasing the mean duration and regulating its random variability to hypoexponential levels. In future research, we plan to modify the distance and position of the enhancer relative to the promoter to better understand the mechanisms of bursting regulation.

We found that one of the main contributors to the formation of the gene expression pattern is activity time (Figure 5), and there is also a weak dependence of the mean $\langle \tau_{OFF} \rangle$ on transcription activity. This observation is similar to previous findings [85, 87]. The activity time and $\langle \tau_{OFF} \rangle$ are related variables and their regulation can share similar mechanisms of action. Specifically, we found cells with isolated periods of long inactivity (>12 min) in between bursts suggesting that these cells can have more than one activity time. In recent work [88], monitoring the transcriptional activity of a promoter under the control of Polycomb proteins has shown a switch between an OFF state and a transcriptionally *permissive* state, with multiple bursts occurring during the permissive period. In this context, the activity time can be understood as the permissive period that is developmentally regulated to drive spatial expression patterning.

The observed importance of activity time in controlling gene expression patterns raises interesting questions about its molecular regulation. Since activity time represents the temporal window during which a gene undergoes bursting cycles, it probably reflects the duration of allowed promoter states for transcriptional activation. We hypothesize that binding of transcription factors (TFs) to cis-regulatory sequences could control this window of activity. For example, Dorsal, a transcription factor that activates *sna* and *rho* studied here, shows a graded expression along the dorsoventral axis of the embryo, showing its highest expression at the most ventral point and a gradual decrease in expression level [89]. The activity time of the Dorsal target genes (*sna* and *rho*), correlates with Dorsal level. Specifically, Dorsal target genes show a longer activity time in the ventral region compared to the more dorsolateral domain. Moreover, cooperative binding of multiple TFs at enhancers may create a stable microenvironment that persists for the duration of the activity time. Our observation that different enhancers (such as *snaPE* versus *snaSE*) exhibit distinct activity time distributions supports this model, as these enhancers contain different combinations of TF binding sites that could affect the stability and duration of active transcriptional states.

In summary, we present a simplified model to capture the stochastic dynamics of transcription in single cells and propose high-throughput techniques to infer promoter states. Our results reveal that these burst timing statistics exhibit uniform spatial patterns within the embryo, with ON times showing tighter statistical fluctuations compared to the OFF times. Moreover, a major determinant of spatial expression regulation for different developmental enhancers is the activity time that is related to the permissive period of transcription.

Methods

Experimental details

MS2/MCP system is used to visualize nascent transcripts (Figure 1C). To build the MS2/MCP mechanism, 24 copies of MS2 sequences are inserted into the 5' UTR or 3' UTR of the target gene of interest. Upon transcription, MS2 sequences form a stem-loop structure, which can bind to maternally loaded MCP proteins fused with GFP. As a result, clustered MCP-GFP molecules at the site of active transcription can be detected as fluorescent puncta in each nucleus (Figure 1B).

The data analyzed in this article, for *rho*, *Kr*, and *sna* genes, correspond to the experiments carried out by the reference [40], where the regulatory enhancer of each gene drives the expression of the *MS2-yellow* reporter gene (Figure 1C). For *eve*, we used data from [41], where endogenous

eve is tagged with MS2 using CRISPR-mediated genome editing. For the nonendogenous data, it has been observed that the dynamics of the reporter gene recapitulate the expression of the endogenous gene. All images are taken in embryos 2-3 hours after fertilization, corresponding to the 14th nuclear cycle (NC14). From each live imaging movie, we track the fluorescence intensity of the MS2 puncta in each nucleus for the duration of NC14 and use it as a measure of instantaneous transcriptional activity (Figure 1D [↗](#)).

Transcription state Inference

As shown in Figure 1D [↗](#), a typical fluorescence trajectory reveals complex dynamics. We can segment the complete signal into periods of no activity when the fluorescence stays at a basal level (before the first burst and after the last one) and periods of activity when transcriptional bursting occurs. Our goal is to determine the precise moments at which these bursts occur which are related to the moment in which the fluorescence signal rises. To do so, we first explain the model on which the inference method is based and then explain the method itself.

Phenomenological modeling of the fluorescence signal dynamics

To simplify our analysis, we consider that the fluorescence signal dynamics is driven mainly by the transcriptional state. This state exists in two modes: ON and OFF. When the transcriptional state is ON, high transcriptional activity results in rising fluorescence levels (illustrated by dark green periods in Figure 1D [↗](#)). Conversely, during the OFF state, the absence of RNA polymerase recruitment causes the fluorescence signal to decrease or stabilize at its base level (light green periods in Figure 1D [↗](#)).

Given the possible transcriptional states (ON and OFF), we define the function $\phi(t)$ as follows:

$$\phi(t) = \begin{cases} 1 & \text{if transcription state is ON} \\ 0 & \text{if transcription state is OFF,} \end{cases} \quad (1)$$

which quantifies the transcriptional state value. When the state is ON, we assume that the fluorescence levels increase at a constant rate. Similarly, when the gene is OFF, the fluorescence also decreases at a constant rate. This linear accumulation and decay approximation was concluded after a detailed analysis of fluorescence accumulation and decay trajectories in response to ON/OFF states, as presented in Figure S1. In addition to a good fit with linear functions of time, we observe that the slopes for decreasing/increasing dynamics have very similar mean absolute values (Figure S2).

In this context, we simplify the dynamics of the fluorescence signal assuming that the signal increases and decreases with the same absolute rate λ that could be different for each cell. Given the transcriptional state $\phi(t)$, the fluorescence signal y^m is modeled as per the following ordinary differential equation:

$$\frac{dy^m}{dt} = \begin{cases} \lambda(2\phi(t) - 1); & \text{if } y^m > y^\dagger \\ 0; & \text{otherwise,} \end{cases} \quad (2)$$

with the superscript m standing for the *model-predicted* signal. The burst threshold $y^\dagger$ is defined such that below this threshold, we assume that no bursting occurs and the transcription state is OFF (Figure 1 [↗](#)). To determine $y^\dagger$, we selected the maximum fluorescence level in cells that did not exhibit noticeable bursts. This value was found to be $y^\dagger = 0.1$ (a.u.). Since basal fluorescence levels did not differ significantly between genes, we used the same threshold for all genes. Next, we will explain how, assuming (2), we infer the most probable λ and the corresponding transcriptional state for each cell.

Transcription state inference method

Since the data consists of sampling the fluorescence signal every $\Delta t = 20$ s, we consider a discrete-time analog of (2) that follows

$$y_{t+\Delta t}^m = \begin{cases} y_t^m + (\lambda \Delta t)(2\phi_t - 1); & \text{if } y_t^m > y^\dagger \\ y_t^m; & \text{otherwise,} \end{cases} \quad (3)$$

where $t = \{0, \Delta t, 2\Delta t, \dots\}$. Given experimental measurements y_t , our objective is to infer the transcriptional state ϕ_t that best explains the data. The technical details of the defined algorithm are presented in Supplementary Information Section S3, and are briefly summarized here.

The inference method starts with the experimental trajectory y_t , which is first smoothed to obtain the signal $\tilde{y}_t$ by convolving it with a triangular kernel function to not flatten the signal peaks too much (see Supplementary Information Section S3). Next, we propose a value for λ for all transcriptional bursts in a single nucleus. Given $\tilde{y}_t$ and λ , we first obtain the *transcription activity* ρ_t defined as

$$\rho_t := \begin{cases} \frac{1}{2}(\tilde{y}_{t+\Delta t} - \tilde{y}_t)/(\lambda \Delta t) + 1; & \text{if } \tilde{y}_t > y^\dagger \\ \frac{1}{2}(\tilde{y}_{t+\Delta t} - \tilde{y}_t)/(\lambda \Delta t); & \text{otherwise,} \end{cases} \quad (4)$$

and is a quantity which estimates the rate of signal increase relative to λ . We perform a new round of smoothing of ρ_t with a square kernel function (see Supplementary Information section 3) obtaining the smoothed $\tilde{\rho}_t$. In this way, the inferred transcriptional state ϕ_t is calculated as follows:

$$\phi_t := \begin{cases} 1; & \text{if } \tilde{\rho}_t > 0.5 \\ 0; & \text{otherwise.} \end{cases} \quad (5)$$

The resulting ϕ_t is replaced in (3) obtaining a trajectory y_t^m that depends parametrically on the particular λ . Finally, the optimal λ , denoted as λ^* , is defined as the loading rate for that specific nuclei that minimizes the mean square distance between the data y_t and the resultant y_t^m defined as:

$$\lambda^* = \underset{\lambda}{\operatorname{argmin}} \left(\sum_t (y_t^m(\lambda) - y_t)^2 \right). \quad (6)$$

The optimal λ^* will predict the optimal ϕ_t by using (5) which is taken as the most accurate trajectory of the transcriptional state. Examples of best-fit trajectories ϕ_t and y_t^m are shown in Figure S3D-E, respectively.

Data accessibility statement

Data set and data processing scripts are publicly available at <https://doi.org/10.5281/zenodo.13208281> [90].

Acknowledgements

B.L. is funded by NIH R35 GM133425. A.S. and C.N. acknowledge support from NIH-NIGMS through grant R35GM148351.

Additional information

Author Contributions

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- **Data Curation:** B.L., Z.V., C.N.
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- **Software:** C.N., Z.V.
- **Supervision:** A.S., B.L.
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Funding

National Institutes of Health (R35GM133425)

National Institutes of Health (R35GM148351)

Additional files

Supplementary Information [↗](#)

References

- [1] Halder Georg, Callaerts Patrick, Gehring Walter J (1995) **Induction of ectopic eyes by targeted expression of the eyeless gene in Drosophila** *Science* **267**:1788–1792 [Google Scholar](#)
- [2] Riedel-Kruse Ingmar H, Muller Claudia, Oates Andrew C (2007) **Synchrony dynamics during initiation, failure, and rescue of the segmentation clock** *Science* **317**:1911–1915 [Google Scholar](#)
- [3] Dubuis Julien O, Samanta Reba, Gregor Thomas (2013) **Accurate measurements of dynamics and reproducibility in small genetic networks** *Molecular Systems Biology* **9**:639 [Google Scholar](#)
- [4] Sakabe Noboru Jo, Savic Daniel, Nobrega Marcelo A (2012) **Transcriptional enhancers in development and disease** *Genome Biology* **13**:1–11 [Google Scholar](#)
- [5] Claringbould Annique, Zaugg Judith B (2021) **Enhancers in disease: molecular basis and emerging treatment strategies** *Trends in Molecular Medicine* **27**:1060–1073 [Google Scholar](#)
- [6] Elowitz M. B., Levine A. J., Siggia E. D., Swain P. S. (2002) **Stochastic gene expression in a single cell** *Science* **297**:1183–1186 [Google Scholar](#)
- [7] Raj Arjun, Van Oudenaarden Alexander (2008) **Nature, nurture, or chance: stochastic gene expression and its consequences** *Cell* **135**:216–226 [Google Scholar](#)
- [8] Holmes William R, de Mochel Nabora Soledad Reyes, Wang Qixuan, Du Huijing, Peng Tao, Chiang Michael, Cinquin Olivier, Cho Ken, Nie Qing (2017) **Gene expression noise enhances robust organization of the early mammalian blastocyst** *PLOS Computational Biology* **13**:e1005320 [Google Scholar](#)
- [9] Voortman Lukas, Jr Robert J Johnston (2022) **Transcriptional repression in stochastic gene expression, patterning, and cell fate specification** *Developmental Biology* **481**:129–138 [Google Scholar](#)
- [10] Araújo Ilka Schultheiß, Pietsch Jessica Magdalena, Keizer Emma Mathilde, Greese Bettina, Balkunde Rachappa, Fleck Christian, Hülkamp Martin (2017) **Stochastic gene expression in arabidopsis thaliana** *Nature Communications* **8**:2132 [Google Scholar](#)
- [11] Ferraro Teresa, Esposito Emilia, Mancini Laure, Ng Sam, Lucas Tanguy, Coppey Mathieu, Dostatni Nathalie, Walczak Aleksandra M, Levine Michael, Lagha Mounia (2016) **Transcriptional memory in the Drosophila embryo** *Current Biology* **26**:212–218 [Google Scholar](#)
- [12] Zhang Lei, Radtke Kelly, Zheng Likun, Cai Anna Q, Schilling Thomas F, Nie Qing (2012) **Noise drives sharpening of gene expression boundaries in the zebrafish hindbrain** *Molecular Systems Biology* **8**:613 [Google Scholar](#)
- [13] Desai Ravi V, Chen Xinyue, Martin Benjamin, Chaturvedi Sonali, Hwang Dong Woo, Li Weihan, Yu Chen, Ding Sheng, Thomson Matt, Singer Robert H, et al. (2021) **A DNA repair pathway can**

regulate transcriptional noise to promote cell fate transitions *Science* **373**:eabc6506 [Google Scholar](#)

- [14] Keskin Sevdenur, Devakanmalai Gnanapackiam S, Bin Kwon Soo, Vu Ha T, Hong Qiyuan, Lee Yin Yeng, Soltani Mohammad, Singh Abhyudai, Ay Ahmet, rul Ertuğ, Özbudak M (2018) **Noise in the vertebrate segmentation clock is boosted by time delays but tamed by notch signaling** *Cell Reports* **23**:2175–2185 [Google Scholar](#)
- [15] Zinani Oriana QH, lu Kemal Keseroğ, Dey Supravat, Ay Ahmet, Singh Abhyudai, rul Ertuğ, Özbudak M (2022) **Gene copy number and negative feedback differentially regulate transcriptional variability of segmentation clock genes** *iScience* **25** [Google Scholar](#)
- [16] Raj A., Peskin C.S., Tranchina D., Vargas D.Y., Tyagi S. (2006) **Stochastic mRNA synthesis in mammalian cells** *PLOS Biology* **4**:e309 [Google Scholar](#)
- [17] Ramsköld Daniel, Hendriks Gert-Jan, Larsson Anton JM, Mayr Juliane V, Ziegenhain Christoph, Hagemann-Jensen Michael, Hartmanis Leonard, Sandberg Rickard (2024) **Single-cell new RNA sequencing reveals principles of transcription at the resolution of individual bursts** *Nature Cell Biology* **26**:1725–1733 [Google Scholar](#)
- [18] Jeziorska DM, Tunnacliffe EAJ, Brown JM, Ayyub H, Sloane-Stanley J, Sharpe JA, Lagerholm BC, Babbs C, Smith AJH, Buckle VJ, et al. (2022) **On-microscope staging of live cells reveals changes in the dynamics of transcriptional bursting during differentiation** *Nature Communications* **13**:6641 [Google Scholar](#)
- [19] Wan Yihan, Anastasakis Dimitrios G, Rodriguez Joseph, Palangat Murali, Gudla Prabhakar, Zaki George, Tandon Mayank, Pegoraro Gianluca, Chow Carson C, Hafner Markus, et al. (2021) **Dynamic imaging of nascent RNA reveals general principles of transcription dynamics and stochastic splice site selection** *Cell* **184**:2878–2895 [Google Scholar](#)
- [20] Porello Emilia A Leyes, Trudeau Robert T, Lim Bomyi (2023) **Transcriptional bursting: stochasticity in deterministic development** *Development* **150**:dev201546 [Google Scholar](#)
- [21] Fukaya Takashi (2023) **Enhancer dynamics: Unraveling the mechanism of transcriptional bursting** *Science Advances* **9**:eadj3366 [Google Scholar](#)
- [22] Dar Roy D., Shaffer Sydney M., Singh Abhyudai, Razooky Brandon S., Simpson Michael L., Raj Arjun, Weinberger Leor S. (2016) **Transcriptional bursting explains the noise-versus-mean relationship in mRNA and protein levels** *PLOS One* **11** [Google Scholar](#)
- [23] Corrigan Adam M., Chubbemail Jonathan R. (2014) **Regulation of transcriptional bursting by a naturally oscillating signal** *Current Biology* **24**:205–211 [Google Scholar](#)
- [24] Ochiai Hiroshi, Hayashi Tetsutaro, Umeda Mana, Yoshimura Mika, Harada Akihito, Shimizu Yukiko, Nakano Kenta, Saitoh Noriko, Liu Zhe, Yamamoto Takashi, et al. (2020) **Genome-wide kinetic properties of transcriptional bursting in mouse embryonic stem cells** *Science Advances* **6** [Google Scholar](#)
- [25] Larson Daniel R, Fritsch Christoph, Sun Liang, Meng Xiuhau, Lawrence David S, Singer Robert H (2013) **Direct observation of frequency modulated transcription in single cells using light activation** *eLife* **2** [Google Scholar](#)
- [26] Cavallaro Massimo, Walsh Mark D, Jones Matt, Teahan James, Tiberi Simone, Finkenstädt Bärbel, Heben-streit Daniel (2021) **3'-5' crosstalk contributes to transcriptional bursting**

- [27] Yunger Sharon, Rosenfeld Liat, Garini Yuval, Shav-Tal Yaron (2010) **Single-allele analysis of transcription kinetics in living mammalian cells** *Nature Methods* 7:631–633 [Google Scholar](#)
- [28] Larson Daniel R, Zenklusen Daniel, Wu Bin, Chao Jeffrey A, Singer Robert H (2011) **Real-time observation of transcription initiation and elongation on an endogenous yeast gene** *Science* 332:475–478 [Google Scholar](#)
- [29] Corrigan Adam M, Tunnaclyffe Edward, Cannon Danielle, Chubb Jonathan R (2016) **A continuum model of transcriptional bursting** *eLife* 5 [Google Scholar](#)
- [30] Bothma Jacques P, Garcia Hernan G, Esposito Emilia, Schlissel Gavin, Gregor Thomas, Levine Michael (2014) **Dynamic regulation of eve stripe 2 expression reveals transcriptional bursts in living Drosophila embryos** *Proceedings of the National Academy of Sciences* 111:10598–10603 [Google Scholar](#)
- [31] Dar Roy D, Razooky Brandon S, Singh Abhyudai, Trimeloni Thomas V, McCollum James M, Cox Chris D, Simpson Michael L, Weinberger Leor S (2012) **Transcriptional burst frequency and burst size are equally modulated across the human genome** *Proceedings of the National Academy of Sciences* 109:17454–17459 [Google Scholar](#)
- [32] Padovan-Merhar Olivia, Nair Gautham P, Bialesch Andrew G, Mayer Andreas, Scarfone Steven, Foley Shawn W, Wu Angela R, Churchman L Stirling, Singh Abhyudai, Raj Arjun (2015) **Single mammalian cells compensate for differences in cellular volume and DNA copy number through independent global transcriptional mechanisms** *Molecular Cell* 58:339–352 [Google Scholar](#)
- [33] Chong Shasha, Chen Chongyi, Ge Hao, Xie X. Sunney (2014) **Mechanism of transcriptional bursting in bacteria** *Cell* 158:314–326 [Google Scholar](#)
- [34] Singh A., Razooky Brandon, Cox Chris D., Simpson Michael L., Weinberger Leor S. (2010) **Transcriptional bursting from the HIV-1 promoter is a significant source of stochastic noise in HIV-1 gene expression** *Biophysical Journal* 98:L32–L34 [Google Scholar](#)
- [35] Schuh Lea, Saint-Antoine Michael, Sanford Eric M, Emert Benjamin L, Singh Abhyudai, Marr Carsten, Raj Arjun, Goyal Yogesh (2020) **Gene networks with transcriptional bursting recapitulate rare transient coordinated high expression states in cancer** *Cell Systems* 10:363–378 [Google Scholar](#)
- [36] Singh A., Razooky B. S., Dar R. D., Weinberger L. S. (2012) **Dynamics of protein noise can distinguish between alternate sources of gene-expression variability** *Molecular Systems Biology* 8 [Google Scholar](#)
- [37] Mold Jeff E, Weissman Martin H, Ratz Michael, Hagemann-Jensen Michael, Hård Joanna, Eriksson Carl-Johan, Toosi Hosein, Berghensträhle Joseph, Ziegenhain Christoph, von Berlin Leonie, et al. (2024) **Clonally heritable gene expression imparts a layer of diversity within cell types** *Cell Systems* 15:149–165 [Google Scholar](#)
- [38] Chubb Jonathan R, Trcek Tatjana, Shenoy Shailesh M, Singer Robert H (2006) **Transcriptional pulsing of a developmental gene** *Current Biology* 16:1018–1025 [Google Scholar](#)
- [39] Golding Ido, Paulsson Johan, Zawilski Scott M, Cox Edward C (2005) **Real-time kinetics of gene activity in individual bacteria** *Cell* 123:1025–1036 [Google Scholar](#)

- [40] Fukaya Takashi, Lim Bomyi, Levine Michael (2016) **Enhancer control of transcriptional bursting** *Cell* **166**:358–368 [Google Scholar](#)
- [41] Lim Bomyi, Fukaya Takashi, Heist Tyler, Levine Michael (2018) **Temporal dynamics of pair-rule stripes in living *Drosophila* embryos** *Proceedings of the National Academy of Sciences* **115**:8376–8381 [Google Scholar](#)
- [42] Bertrand Edouard, Chartrand Pascal, Schaefer Matthias, Shenoy Shailesh M, Singer Robert H, Long Roy M (1998) **Localization of ash1 mRNA particles in living yeast** *Molecular Cell* **2**:437–445 [Google Scholar](#)
- [43] Garcia Hernan G, Tikhonov Mikhail, Lin Albert, Gregor Thomas (2013) **Quantitative imaging of transcription in living *Drosophila* embryos links polymerase activity to patterning** *Current Biology* **23**:2140–2145 [Google Scholar](#)
- [44] Herbert Jächle, Hoch Michael, Ankratz Michael J, Gerwin Nicole, Sauer Frank, Brönner Günter (1992) **Transcriptional control by *Drosophila* gap genes** *Journal of Cell Science* **1992**:39–51 [Google Scholar](#)
- [45] Ip Y Tony, Park Ronald E, Kosman David, Bier Ethan, Levine Michael (1992) **The dorsal gradient morphogen regulates stripes of rhomboid expression in the presumptive neuroectoderm of the *Drosophila* embryo** *Genes & Development* **6**:1728–1739 [Google Scholar](#)
- [46] Perry Michael W, Boettiger Alistair N, Bothma Jacques P, Levine Michael (2010) **Shadow enhancers foster robustness of *Drosophila* gastrulation** *Current Biology* **20**:1562–1567 [Google Scholar](#)
- [47] Goto Tadaatsu, Macdonald Paul, Maniatis Tom (1989) **Early and late periodic patterns of even-skipped expression are controlled by distinct regulatory elements that respond to different spatial cues** *Cell* **57**:413–422 [Google Scholar](#)
- [48] Berrocal Augusto, Lammers Nicholas C, Garcia Hernan G, Eisen Michael B (2020) **Kinetic sculpting of the seven stripes of the *Drosophila* even-skipped gene** *eLife* **9** [Google Scholar](#)
- [49] Ingham PW (1988) **The molecular genetics of embryonic pattern formation in *Drosophila*** *Nature* **335**:25–34 [Google Scholar](#)
- [50] Hoch Michael, Schröder C, Seifert Eveline, Jäckle H (1990) **cis-acting control elements for *krüppel* expression in the *Drosophila* embryo** *The EMBO journal* **9**:2587–2595 [Google Scholar](#)
- [51] Mahat Dig B, Tippens Nathaniel D, Martin-Rufino Jorge D, Waterton Sean K, Fu Jiayu, Blatt Sarah E, Sharp Phillip A (2024) **Single-cell nascent RNA sequencing unveils coordinated global transcription** *Nature* :1–8 [Google Scholar](#)
- [52] Bothma Jacques P, Garcia Hernan G, Ng Samuel, Perry Michael W, Gregor Thomas, Levine Michael (2015) **Enhancer additivity and non-additivity are determined by enhancer strength in the *Drosophila* embryo** *eLife* **4** [Google Scholar](#)
- [53] Pankratz Michael J, Jäckle Herbert (1990) **Making stripes in the *Drosophila* embryo** *Trends in Genetics* **6**:287–292 [Google Scholar](#)
- [54] Berrocal Augusto, Lammers Nicholas C, Garcia Hernan G, Eisen Michael B (2024) **Unified bursting strategies in ectopic and endogenous even-skipped expression patterns** *eLife* **12** [Google Scholar](#)

- [55] Gabriele Michele, Brandão Hugo B, Grosse-Holz Simon, Jha Asmita, Dailey Gina M, Cattoglio Claudia, Hsieh Tsung-Han S, Mirny Leonid, Zechner Christoph, Hansen Anders S (2022) **Dynamics of ctf-and cohesin-mediated chromatin looping revealed by live-cell imaging** *Science* **376**:496–501 [Google Scholar](#)
- [56] Fraser LaTasha CR, Dikdan Ryan J, Dey Supravat, Singh Abhyudai, Tyagi Sanjay (2021) **Reduction in gene expression noise by targeted increase in accessibility at gene loci** *Proceedings of the National Academy of Sciences* **118**:e2018640118 [Google Scholar](#)
- [57] May Dennis, Yun Sangwon, Gonzalez David G, Park Sangbum, Chen Yanbo, Lathrop Elizabeth, Cai Biao, Xin Tianchi, Zhao Hongyu, Wang Siyuan, et al. (2023) **Live imaging reveals chromatin compaction transitions and dynamic transcriptional bursting during stem cell differentiation in vivo** *eLife* **12** [Google Scholar](#)
- [58] Kenworthy Charles A, Haque Nayem, Liou Shu-Hao, Chandris Panagiotis, Wong Vincent, Dziuba Patrycja, Lavis Luke D, Liu Wei-Li, Singer Robert H, Coleman Robert A (2022) **Bromodomains regulate dynamic targeting of the pbaf chromatin-remodeling complex to chromatin hubs** *Biophysical Journal* **121**:1738–1752 [Google Scholar](#)
- [59] Brouwer Ineke, Kerklingh Emma, van Leeuwen Fred, Lenstra Tineke L (2023) **Dynamic epistasis analysis reveals how chromatin remodeling regulates transcriptional bursting** *Nature structural & molecular biology* **30**:692–702 [Google Scholar](#)
- [60] Bohrer Christopher H, Larson Daniel R (2023) **Synthetic analysis of chromatin tracing and live-cell imaging indicates pervasive spatial coupling between genes** *eLife* **12** [Google Scholar](#)
- [61] Bartman Caroline R., Hsu Sarah C., Hsiung Chris C.-S., Raj Arjun, Blobel Gerd A. (2016) **Enhancer regulation of transcriptional bursting parameters revealed by forced chromatin looping** *Molecular Cell* **62**:237–247 [Google Scholar](#)
- [62] Robles-Rebollo Irene, Cuartero Sergi, Canellas-Socias Adria, Wells Sarah, Karimi Mohammad M, Mereu Elisabetta, Chivu Alexandra G, Heyn Holger, Whilding Chad, Dormann Dirk, et al. (2022) **Cohesin couples transcriptional bursting probabilities of inducible enhancers and promoters** *Nature Communications* **13**:4342 [Google Scholar](#)
- [63] Du Manyu, Stitzinger Simon Hendrik, Spille Jan-Hendrik, Cho Won-Ki, Lee Choongman, Hijaz Mohammed, Quintana Andrea, Cissé Ibrahim I (2024) **Direct observation of a condensate effect on super-enhancer controlled gene bursting** *Cell* **187**:331–344 [Google Scholar](#)
- [64] Keller Samuel H, Jena Siddhartha G, Yamazaki Yuji, Lim Bomyi (2020) **Regulation of spatiotemporal limits of developmental gene expression via enhancer grammar** *Proceedings of the National Academy of Sciences* **117**:15096–15103 [Google Scholar](#)
- [65] Syed Sahla, Duan Yifei, Lim Bomyi (2023) **Modulation of protein-dna binding reveals mechanisms of spatiotemporal gene control in early Drosophila embryos** *eLife* **12** [Google Scholar](#)
- [66] Dobrinić Paula, Szczurek Aleksander T, Klose Robert J (2021) **Prc1 drives polycomb-mediated gene repression by controlling transcription initiation and burst frequency** *Nature Structural & Molecular Biology* **28**:811–824 [Google Scholar](#)
- [67] Ma Liang, Gao Zeyue, Wu Jiegen, Zhong Bijunyao, Xie Yuchen, Huang Wen, Lin Yihan (2021) **Co-condensation between transcription factor and coactivator p300 modulates**

transcriptional bursting kinetics *Molecular Cell* **81**:1682–1697 [Google Scholar](#)

- [68] Fountas Charis, Lenstra Tineke L (2024) **Better together: how cooperativity influences transcriptional bursting** *Current Opinion in Genetics & Development* **89** [Google Scholar](#)
- [69] Tantale Katjana, Garcia-Oliver Encar, Robert Cécile, LÕhostis Adèle, Yang Yueyuxiao, Tsanov Nikolay, Topno Rachel, Gostan Thierry, Kozulic-Pirher Alja, Basu-Shrivastava Meenakshi, et al. (2021) **Stochastic pausing at latent hiv-1 promoters generates transcriptional bursting** *Nature Communications* **12**:4503 [Google Scholar](#)
- [70] Pimmett Virginia L, Dejean Matthieu, Fernandez Carola, Trullo Antonio, Bertrand Edouard, Radulescu Ovidiu, Lagha Mounia (2021) **Quantitative imaging of transcription in living Drosophila embryos reveals the impact of core promoter motifs on promoter state dynamics** *Nature Communications* **12**:4504 [Google Scholar](#)
- [71] Suter David M, Molina Nacho, Gatfield David, Schneider Kim, Schibler Ueli, Naef Felix (2011) **Mammalian genes are transcribed with widely different bursting kinetics** *Science* **332**:472–474 [Google Scholar](#)
- [72] Choubey Sandeep, Kondev Jane, Sanchez Alvaro (2015) **Deciphering transcriptional dynamics in vivo by counting nascent RNA molecules** *PLOS Computational Biology* **11**:e1004345 [Google Scholar](#)
- [73] Tantale Katjana, Mueller Florian, Kozulic-Pirher Alja, Lesne Annick, Victor Jean-Marc, Robert Marie-Cécile, Capozzi Serena, Chouaib Racha, Bäcker Volker, Mateos-Langerak Julio, et al. (2016) **A single-molecule view of transcription reveals convoys of RNA polymerases and multi-scale bursting** *Nature Communications* **7**:12248 [Google Scholar](#)
- [74] Xu Zikai, Soltani Mohammad, Singh Abhyudai (2018) **Exact statistical moments of multi-mode stochastic hybrid systems with renewal transitions** In: *2018 IEEE Conference on Decision and Control (CDC)* IEEE pp. 3510–3515 [Google Scholar](#)
- [75] Jia Tao, Kulkarni Rahul V. (2011) **Intrinsic noise in stochastic models of gene expression with molecular memory and bursting** *Journal of Mathematical Biology* **106** [Google Scholar](#)
- [76] Wu Renjie, Zhou Bangyan, Wang Wei, Liu Feng (2023) **Regulatory mechanisms for transcriptional bursting revealed by an event-based model** *Research* **6** [Google Scholar](#)
- [77] Shi Changhong, Yang Xiyan, Zhang Jiajun, Zhou Tianshou (2023) **Stochastic modeling of the mrna life process: A generalized master equation** *Biophysical Journal* **122**:4023–4041 [Google Scholar](#)
- [78] Shi Changhong, Yang Xiyan, Zhou Tianshou, Zhang Jiajun (2024) **Nascent RNA kinetics with complex promoter architecture: Analytic results and parameter inference** *Physical Review E* **110**:034413 [Google Scholar](#)
- [79] Chen Meiling, Luo Songhao, Cao Mengfang, Guo Chengjun, Zhou Tianshou, Zhang Jiajun (2022) **Exact distributions for stochastic gene expression models with arbitrary promoter architecture and translational bursting** *Physical Review E* **105**:014405 [Google Scholar](#)
- [80] Stinchcombe Adam R, Peskin Charles S, Tranchina Daniel (2012) **Population density approach for discrete mrna distributions in generalized switching models for stochastic gene expression** *Physical Review E—Statistical, Nonlinear, and Soft Matter Physics* **85**:061919 [Google Scholar](#)

- [81] Kumar Niraj, Singh Abhyudai, Kulkarni Rahul V. (2015) **Transcriptional bursting in gene expression: analytical results for general stochastic models** *PLOS Computational Biology* **11** [Google Scholar](#)
- [82] Vahdat Zahra, Singh Abhyudai (2024) **Quantifying statistics of gene product copy-number fluctuations: A stochastic hybrid systems approach** In: *2024 IEEE 63rd Conference on Decision and Control (CDC)* pp. 7792–7797 [Google Scholar](#)
- [83] Cao Zhixing, Filatova Tatiana, Oyarzún Diego A, Grima Ramon (2020) **A stochastic model of gene expression with polymerase recruitment and pause release** *Biophysical Journal* **119**:1002–1014 [Google Scholar](#)
- [84] Singh A (2014) **Transient changes in intercellular protein variability identify sources of noise in gene expression** *Biophysical Journal* **107**:2214–2220 [Google Scholar](#)
- [85] Zoller Benjamin, Little Shawn C, Gregor Thomas (2018) **Diverse spatial expression patterns emerge from unified kinetics of transcriptional bursting** *Cell* **175**:835–847 [Google Scholar](#)
- [86] Beadle Lauren Forbes, Zhou Hongpeng, Rattray Magnus, Ashe Hilary L (2023) **Modulation of transcription burst amplitude underpins dosage compensation in the Drosophila embryo** *Cell Reports* **42** [Google Scholar](#)
- [87] Molina Nacho, Suter David M, Cannavo Rosamaria, Zoller Benjamin, Gotic Ivana, Naef Félix (2013) **Stimulus-induced modulation of transcriptional bursting in a single mammalian gene** *Proceedings of the National Academy of Sciences* **110**:20563–20568 [Google Scholar](#)
- [88] Szczurek Aleksander T, Dimitrova Emilia, Kelley Jessica R, Blackledge Neil P, Klose Robert J (2024) **The polycomb system sustains promoters in a deep off state by limiting pre-initiation complex formation to counteract transcription** *Nature Cell Biology* **26**:1700–1711 [Google Scholar](#)
- [89] Hong Joung-Woo, Hendrix David A, Papatsenko Dmitri, Levine Michael S (2008) **How the dorsal gradient works: insights from postgenome technologies** *Proceedings of the National Academy of Sciences* **105**:20072–20076 [Google Scholar](#)
- [90] Nieto Cesar, Vahdat Zahra (2024) **Data processing pipeline for the article “Analysis of Transcriptional Bursting Dynamics and its Role on Spatial Patterning of D. melanogaster Embryo development”** *Zenodo* <https://doi.org/10.5281/zenodo.13324127>

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Reviewer #1 (Public review):

Summary:

In this article, the authors develop a method to re-analyze published data measuring the transcription dynamics of developmental genes within *Drosophila* embryos. Using a simple framework, they identify periods of transcriptional activity from traces of MS2 signal and analyze several parameters of these traces. In the five data sets they analyzed, the authors find that each transcriptional "burst" has a largely invariant duration, both across spatial positions in the embryo and across different enhancers and genes, while the time between transcriptional bursts varies more. However, they find that the best predictor of the mean transcription levels at different spatial positions in the embryo is the "activity time" -- the total time from the first to the last transcriptional burst in the observed cell cycle.

Strengths:

- (1) The algorithm for analyzing the MS2 transcriptional traces is clearly described and appropriate for the data.
- (2) The analysis of the four transcriptional parameters -- the transcriptional burst duration, the time between bursts, the activity time, and the polymerase loading rate is clearly done and logically explained, allowing the reader to observe the different distributions of these values and the relationship between each of these parameters and the overall expression output in each cell. The authors make a convincing case that the activity time is the best predictor of a cell's expression output.
- (3) The figures are clearly presented and easy to follow.

Weaknesses:

(1) The strength of the relationship between the different transcriptional parameters and the mean expression output is displayed visually in Figures 5 and 7, but is not formally quantified. Given that the tau_off times seem more correlated to mean activity for some enhancers (e.g., rho) than others (e.g., sna SE), the quantification might be useful.

(2) There are some mechanistic details that are not discussed in depth. For example, the authors observe that the accumulation and degradation of the MS2 signal have similar slopes. However, given that the accumulation represents the transcription of MS2 loops, while the degradation represents diffusion of nascent transcripts away from the site of transcription, there is no mechanistic expectation for this. The degradation of signal seems likely to be a property of the mRNA itself, which shouldn't vary between cells or enhancer reporters, but the accumulation rate may be cell- or enhancer-specific. Similarly, the activity time depends both on the time of transcription onset and the time of transcription cessation. These two processes may be controlled by different transcription factor properties or levels and may be interesting to disentangle.

(3) There are previous analyses of the eve stripe dynamics, which the authors cite, but do not compare the results of their work to the previous work in depth.

<https://doi.org/10.7554/eLife.107610.1.sa3>

Reviewer #2 (Public review):

Summary:

In this work, Nieto et al. investigate how spatial gene expression patterns in the early *Drosophila* embryo are regulated at the level of transcriptional bursting. Using live-cell MS2 imaging data of four reporter constructs and the endogenous *eve* gene, the authors extract temporal dynamics of nascent transcription at single-cell resolution. They implement a novel, simplified algorithm to infer promoter ON/OFF states based on fluorescence slope dynamics and use this to quantify burst duration (Ton), inter-burst duration (Toff), and total activity time across space.

The key finding is that while Ton and Toff remain relatively constant across space, the activity time—the window between first and last burst—is spatially modulated and best explains mean expression differences across the embryo. This uncovers a general strategy where early embryonic patterning genes modulate the duration of their transcriptionally permissive states, rather than the frequency or strength of bursting itself. The manuscript also shows that different enhancers of the same gene (e.g., *sna* proximal vs. shadow) can differentially modulate Toff and activity time, providing mechanistic insight into enhancer function.

Strengths:

The manuscript introduces activity time as a major, previously underappreciated determinant of spatial gene expression, distinct from Ton and Toff, providing an intuitive mechanistic link between temporal bursting and spatial patterning.

The authors develop a tractable inference algorithm based on linear accumulation/decay rates of MS2 fluorescence, allowing efficient burst state segmentation across thousands of trajectories.

Analysis across multiple biological replicates and different genes/enhancers lends confidence to the reproducibility and generalizability of the findings.

By analyzing both synthetic reporter constructs and an endogenous gene (*eve*), the work provides a coherent view of how enhancer architecture and spatial regulation are intertwined with transcriptional kinetics.

The supplementary information extends the biological findings with a gene expression noise model that accounts for non-exponential dwell times and illustrates how low-variability Ton buffers stochasticity in transcript levels.

Weaknesses:

The manuscript does not clearly delineate how this analysis extends beyond the prior landmark study (citation #40: Fukaya et al., 2016). While the current manuscript offers new modeling and statistics, more explicit clarification of what is novel in terms of biological conclusions and methodological advancement would help position the work.

While the methods are explained in detail in the Supplementary Information, the manuscript would benefit from including a diagrammatic model and explicitly clarifying whether the model is descriptive or predictive in scope.

The interpretation that fluorescence decay reflects RNA degradation could be confounded by polymerase runoff or transcript diffusion from the transcription site. These potential limitations are not thoroughly discussed.

The so-called loading rate is used as an empirical parameter in fitting fluorescence traces, but is not convincingly linked to distinct biological processes. The manuscript would benefit from a more precise definition or reframing of this term.

Impact and Utility:

The study provides a general and scalable framework for dissecting transcriptional kinetics in developing embryos, with implications for understanding enhancer logic and developmental robustness. The algorithm is suitable for adaptation to other live-imaging datasets and could be useful across systems where temporal transcriptional variability is being quantified. By highlighting activity time as a key regulatory axis, the work shifts attention to transcriptionally permissive windows as a primary developmental control layer.

This work will be of interest to: developmental biologists investigating spatial gene expression, researchers studying transcriptional regulation and noise, quantitative biologists developing models for transcriptional dynamics, and imaging and computational biologists working with live single-cell data.

<https://doi.org/10.7554/eLife.107610.1.sa2>

Reviewer #3 (Public review):

Summary:

In this paper, the authors developed a simple algorithm to analyse live imaging transcription data (MS2) and infer various kinetic parameters. They then applied it to analyse data from previous publications on *Drosophila* that measured the dynamics of reporter genes driven by various enhancers alone (*sna*, *Kr*, *rho*), or in an endogenous context (*eve*).

The authors find that the main correlate with mean gene expression levels is the activity time, that is, the time during which the gene is bursting. They also find a correlation with the variation of the off time.

Strengths:

- (1) The findings are very clearly presented.
- (2) The simplicity of the algorithm is nice, and the comparative analysis among the various enhancers can be helpful for the field.

Weaknesses:

- (1) The algorithm is not benchmarked against previously used algorithms in the field to infer ON and OFF times, for example, those based on Hidden Markov models. A comparison would help strengthen the support for this algorithm (if it really works well) or show at which point one must be careful when interpreting this data.
- (2) More broadly, the novelty of the findings and how those fit within the knowledge of the field is not super clear. A better account of previous findings that have already quantified ON, OFF times and so on, and how the current findings fit within those, would help better appreciate the significance of the work.

<https://doi.org/10.7554/eLife.107610.1.sa1>

Author response:

Reviewer #1 (Public review):

(1) The strength of the relationship between the different transcriptional parameters and the mean expression output is displayed visually in Figures 5 and 7, but is not formally quantified. Given that the tau_off times seem more correlated to mean activity for some enhancers (e.g., rho) than others (e.g., sna SE), the quantification might be useful.

We re-plot Figure 5 and Figure 7 to present the correlation between the studied burst parameters. As the reviewer suggested, after quantifying the correlation we can better study the correlation between the cells averaged tau-off and the cell-averaged fluorescence signal in some of the selected enhancers. As a result of these findings we decide to change our message and instead of claiming that the burst statistics are homogeneous over the embryo domain, to claim that these statistics have weak but significant correlations with the cell-averaged mean gene fluorescence.

(2) There are some mechanistic details that are not discussed in depth. For example, the authors observe that the accumulation and degradation of the MS2 signal have similar slopes. However, given that the accumulation represents the transcription of MS2 loops, while the degradation represents diffusion of nascent transcripts away from the site of transcription, there is no mechanistic expectation for this. The degradation of signal seems likely to be a property of the mRNA itself, which shouldn't vary between cells or enhancer reporters, but the accumulation rate may be cell- or enhancer-specific. Similarly, the activity time depends both on the time of transcription onset and the time of transcription cessation. These two processes may be controlled by different transcription factor properties or levels and may be interesting to disentangle.

The accumulation slope represents the rate of nascent transcript production, which depends on transcription initiation frequency and RNA polymerase elongation rate. While transcription initiation rates can vary between enhancers, our results show that the loading rates are relatively comparable across different enhancer sequences (Figure 5D). Instead, the primary difference observed was in activity time and burst frequency, consistent with

previous findings that enhancers predominantly modulate burst frequency (Fukaya et al., 2016). The degradation slope represents the diffusion of completed transcripts away from the transcription site, which should be an intrinsic property of the mRNA molecule and therefore independent of the regulatory sequences driving transcription.

(3) There are previous analyses of the eve stripe dynamics, which the authors cite, but do not compare the results of their work to the previous work in depth.

The goal of this manuscript is to compare transcriptional bursting properties across different enhancers, rather than to provide an in-depth analysis of eve stripe dynamics specifically. We analyzed four transgenic constructs with different enhancers alongside an endogenous eve construct, focusing on comparative bursting parameters rather than detailed eve expression patterns. Additionally, the previously published eve stripe dynamics data came from BAC constructs, whereas our data comes from the endogenous eve locus. This methodological difference makes direct comparison of stripe dynamics less straightforward and less relevant to our central research question about enhancer-driven bursting variability.

Reviewer #2 (Public review):

(1) The manuscript does not clearly delineate how this analysis extends beyond the prior landmark study (citation #40: Fukaya et al., 2016). While the current manuscript offers new modeling and statistics, more explicit clarification of what is novel in terms of biological conclusions and methodological advancement would help position the work.

The prior study (Fukaya et al., 2016) characterized transcriptional bursting qualitatively, focusing on average burst properties per nucleus without systematic mathematical modeling or statistical analysis of burst-to-burst variability. While they demonstrated that enhancer strength correlates with burst frequency, no quantitative framework was developed to dissect the molecular mechanisms underlying these differences or to connect burst dynamics to spatial gene expression patterns.

(1) We developed an explicit mathematical model with rigorous inference algorithms to quantify transcriptional states from fluorescence trajectories; (2) We performed comprehensive statistical analysis of burst timing distributions, revealing that inter-burst intervals follow exponential distributions while burst durations are hypo-exponentially distributed; (3) Most importantly, we discovered that burst kinetics (τ_{ON} , τ_{OFF}) remain remarkably consistent across different genes and spatial locations, while spatial expression gradients arise primarily through modulation of activity time - the temporal window during which bursting occurs. This mechanistic insight reveals that enhancers regulate spatial patterning not by changing intrinsic burst properties, but by controlling the duration of transcriptionally permissive periods.

(2) While the methods are explained in detail in the Supplementary Information, the manuscript would benefit from including a diagrammatic model and explicitly clarifying whether the model is descriptive or predictive in scope.

We plan to prepare the diagrammatic model in the formal response.

(3) The interpretation that fluorescence decay reflects RNA degradation could be confounded by polymerase runoff or transcript diffusion from the transcription site. These potential limitations are not thoroughly discussed. (Write few lines in the discussion)

This concern, related to the interpretation of the predictive model will be addressed in a future work. The decay in the fluorescence signal can be biologically related to the

transcription termination, polymerase detachment, and diffusion. A key limitation of the approach is that the model is phenomenological and does not capture processes that can be addressed with a more mechanistic model.

(4) The so-called loading rate is used as an empirical parameter in fitting fluorescence traces, but is not convincingly linked to distinct biological processes. The manuscript would benefit from a more precise definition or reframing of this term.

We modify the language of our definition of loading rate as follows: “Loading rate is defined as the rate of increase of fluorescence signal following promoter activation. This quantity is a proxy measurement for the rate of RNA Polymerase II transcription initiation.” The full transcription process has multiple mechanisms including chromatin dynamics, 3D enhancer-promoter interactions, transcription factor binding, mRNA polymerase pausing, and interactions between developmental promoter motifs and associated proteins. We did not have access to specific measurements of these mechanisms and therefore cannot provide a solid biological meaning of the model behind the inference algorithm. However, the fact that we have reproducible results in biological replicas can support the robustness of our method at predicting the promoter state in the studied datasets. In the formal response we will compare the performance of our method with other available ones.

Reviewer #3 (Public review):

(1) The algorithm is not benchmarked against previously used algorithms in the field to infer ON and OFF times, for example, those based on Hidden Markov models. A comparison would help strengthen the support for this algorithm (if it really works well) or show at which point one must be careful when interpreting this data.

We are implementing a benchmarking protocol to compare our results with the proposed and already published models. We expect to present this comparison in the formal response.

(2) More broadly, the novelty of the findings and how those fit within the knowledge of the field is not super clear. A better account of previous findings that have already quantified ON, OFF times and so on, and how the current findings fit within those, would help better appreciate the significance of the work.

To have a better clarity of the new findings we modified the title from “Regulation of Transcriptional Bursting and Spatial Patterning in Early Drosophila Embryo Development” to “Temporal Duration of Gene Activity is the main Regulator of Spatial Expression Patterns in Early Drosophila Embryos”.

In short, (1) We developed an explicit mathematical model with rigorous inference algorithms to quantify transcriptional states from fluorescence trajectories; (2) We performed comprehensive statistical analysis of burst timing distributions, revealing that inter-burst intervals follow exponential distributions while burst durations are hypo-exponentially distributed; (3) Most importantly, we discovered that burst kinetics (τ_{ON} , τ_{OFF}) remain remarkably consistent across different genes and spatial locations, while spatial expression gradients arise primarily through modulation of activity time - the temporal window during which bursting occurs. This mechanistic insight reveals that enhancers regulate spatial patterning not by changing intrinsic burst properties, but by controlling the duration of transcriptionally permissive periods.

<https://doi.org/10.7554/eLife.107610.1.sa0>